EVQ-Cosh: Closed-Form Frequency Allocation for Long-Context Extrapolation

Anonymous Author(s)
Affiliation
Address
email

Abstract

Rotary position embedding is often extended at inference time, but its training-time geometric frequency allocation is usually left untouched. We formulate RoPE frequency allocation as a variational inverse problem, derive a broadband surrogate whose stationary density satisfies a second-order ODE, and obtain a closed-form family, EVQ-cosh, in which geometric RoPE is the degenerate $\tau = 0$ limit. Empirically, EVQ beats learnable adaptive positional encoding in DAPE-style extreme extrapolation ($128 \to 8$K: 333.7 vs. 455.3 PPL), showing that positional structure alone can determine extreme long-range behavior. More importantly, EVQ unlocks inference-time scaling: across six seeds, EVQ+YaRN reaches 100% retrieval at 8K, while Geo+YaRN remains at 61% to 65%; in a fair 10% mix comparison, EVQ+YaRN lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K. Under progressive training ($512 \to 1024 \to 2048$), this advantage amplifies superlinearly, extending functional context to 48K ($24\times$ training length) at PPL 2.63. These results identify training-time frequency allocation as a missing foundation for long-context extrapolation.

1 Introduction

Long-context language modeling is often framed as an inference-time scaling problem: if rotary position embedding (RoPE) is compressed, interpolated, or rescaled carefully enough, a model trained at length L_{train} should extrapolate to substantially longer contexts [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024]. This view has produced strong engineering baselines, but it leaves one design choice essentially untouched: the training-time geometric allocation of RoPE frequencies. Standard RoPE treats that allocation as fixed, even though long-range failure is ultimately mediated through frequency collisions in the learned representation.

We take the opposite starting point. Instead of asking only how to rescale positions at inference time, we ask how frequencies should be allocated during training. We formulate RoPE frequency allocation as a variational inverse problem, derive a broadband surrogate in which the stationary density satisfies a second-order ODE, and obtain a closed-form family, EVQ-cosh. In this family, geometric RoPE appears as the degenerate $\tau = 0$ limit rather than a generic optimum.

This change in viewpoint leads to two central empirical findings. First, in the PE-dominant extreme-extrapolation regime, EVQ beats learnable adaptive positional encoding with zero additional parameters: on the DAPE-style $128 \to 8$K setup, EVQ reaches 333.7 PPL while DAPE reaches 455.3. This is the cleanest setting in which positional structure itself is the dominant variable. Second, and more importantly, EVQ acts as the missing training-time foundation for inference-time scaling. Across six seeds, EVQ+YaRN reaches 100% retrieval at 8K, while Geo+YaRN remains at 61% to 65%; in a fair 10% mix comparison, EVQ+YaRN lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

The paper's claim is therefore sharper than "EVQ improves RoPE." Our claim is that standard RoPE underperforms because its geometric frequency allocation is the wrong training-time foundation for long-context extrapolation. Inference-time scaling can only recover what the checkpoint geometry makes available. Once that geometry is improved, the strongest existing inference-time methods become significantly more effective.

This distinction matters operationally. The dominant deployment pattern in long-context LLMs is to keep the training checkpoint fixed and layer stronger inference-time scaling on top. If the checkpoint has already collapsed too much low-frequency resolution into a geometric schedule, then inference-time scaling is trying to stretch positions on top of the wrong substrate. Our thesis is that training-time frequency design is therefore a missing systems primitive, not merely a cleaner mathematical interpretation of RoPE.

Our contributions are:

- We derive a closed-form variational solution for RoPE frequency allocation and show that geometric RoPE is the $\tau = 0$ degenerate case of a broader optimal family.
- We show that closed-form EVQ beats learnable PE in DAPE-style extreme extrapolation, supporting the claim that PE quality itself can determine long-range behavior.
- We demonstrate that the strongest systems result is not merely EVQ over geometric RoPE, but EVQ unlocking YaRN: training-time and inference-time positional optimization are orthogonal and strongly complementary.
- We show that progressive training amplifies EVQ's advantage superlinearly: three-stage context extension ($512 \to 1024 \to 2048$) yields a monotonically growing EVQ+YaRN PPL advantage at 16K ($-34.6\% \to -52.0\% \to -81.2\%$), extending functional context to 48K at PPL 2.63.
- We present a conservative empirical package in which the main claims rely on multi-seed text evidence, while larger-scale single-seed runs and cross-modal transfer results are treated as consistency checks rather than primary anchors.

2 Related Work

Rotary and relative positional methods. Transformers rely on positional structure to recover order information [Vaswani et al., 2017]. Popular alternatives include relative attention biases [Shaw et al., 2018, Raffel et al., 2020], ALiBi [Press et al., 2022], and RoPE / RoFormer [Su et al., 2024]. Our work is closest in object to RoPE, but unlike prior usage it treats frequency allocation itself as the quantity to optimize.

Inference-time context extension. Recent long-context methods often preserve the training checkpoint and modify positional behavior only at inference time, for example via positional interpolation, YaRN, LongRoPE, and related extension heuristics [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024, Anil et al., 2023, Jin et al., 2024]. Survey work already frames positional design as a central obstacle in length extrapolation [Wang et al., 2024]. These methods are practically important and form the relevant systems baseline, but they do not directly optimize the training-time frequency geometry on which extrapolation ultimately relies. Our thesis is not that such methods are unimportant, but that they are operating on a checkpoint whose positional substrate may already be suboptimal.

Learnable and adaptive positional encoding. Another line of work introduces learnable or adaptive positional mechanisms, including Kerple, FIRE, and DAPE [Chi et al., 2023, Li et al., 2024, Zheng et al., 2024]. These methods are the closest comparison point for our PE-dominant experiments because they ask whether adaptive positional structure can improve extrapolation. Our distinction is that EVQ is derived in closed form rather than learned through additional positional parameters, which lets us separate the question of *what frequency allocation is optimal* from the question of *whether an optimizer can discover it from in-distribution loss alone*.

Evaluation for long-context behavior. Long-context evaluation is often split between synthetic retrieval-style diagnostics and broader long-context task suites such as LongBench and RULER [Bai et al., 2024, Hsieh et al., 2024]. Prior work has also shown that long contexts can fail even when

relevant information is technically present [Liu et al., 2024]. We therefore use both PE-dominant extrapolation settings and retrieval-heavy long-context evaluation to separate mechanistic evidence from broader empirical support, and we treat incomplete larger-scale task evidence conservatively.

3 Theory and Method

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency and define $\omega(\phi) = b^{-\phi}$. A discrete RoPE design is therefore equivalent to choosing a density $\rho(\phi)$ and then discretizing it by inverse CDF. Standard geometric RoPE corresponds to a uniform density over ϕ; EVQ instead derives ρ from a variational objective. This reparameterization moves the design space from a hand-written index rule to a continuous allocation problem.

3.2 Exact kernel and broadband surrogate

Under a distance prior $D(\Delta)$, the phase-collision kernel is

$$K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta. \tag{1}$$

Our only approximation step is a broadband surrogate,

$$K_{\mathrm{app}}(\phi_1, \phi_2) \approx \alpha\delta(\phi_1 - \phi_2) + \beta \min(\phi_1, \phi_2), \tag{2}$$

which isolates a diagonal ridge term and a smooth off-diagonal covariance term. The δ term captures local ridge structure, while $\min(\phi_1, \phi_2)$ is the Green kernel of a one-dimensional second-order operator. After this step, the remaining derivation is exact.

3.3 Variational objective and exact ODE

The broadband functional takes the form

$$\mathcal{J}[\rho] = \frac{\alpha}{2} \int \rho(\phi)^2\, d\phi + \frac{\beta}{2} \iint \rho(\phi)\rho(\psi) \min(\phi, \psi)\, d\phi\, d\psi - \mu \int \rho(\phi) b^{-2\phi}\, d\phi. \tag{3}$$

Theorem 1. *After fixing the surrogate K_{app}, the stationary density satisfies*

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}, \tag{4}$$

with a closed-form solution consisting of a hyperbolic tether and a Fisher-driven forcing term.

This is the exact theory backbone used in the paper body: the surrogate is the only approximation, while the ODE and its closed-form family are exact conditional on that surrogate. The homogeneous branch supplies a global tether that redistributes density, while the forcing term preserves local Fisher utility. The practical significance of this decomposition is that it separates two qualitatively different requirements: a global allocation that decides where frequency mass should live, and a local utility term that prevents the solution from collapsing into a purely asymptotic long-range design. EVQ is therefore not an unconstrained long-range distortion of RoPE; it is the stationary compromise induced by the surrogate objective.

3.4 Closed-form EVQ and the geometric limit

The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \qquad u_k = \frac{k}{K}, \tag{5}$$

and inverse frequencies $b^{-\phi_k(\tau)}$.

Theorem 2. *EVQ-cosh converges smoothly to geometric RoPE as $\tau \to 0$. Therefore, geometric RoPE is the $\tau = 0$ degenerate limit of the EVQ family rather than a generic optimum.*

The discrete implementation then follows by evaluating $\phi_k(\tau)$ at evenly spaced quantiles u_k. In this form, EVQ changes no architecture component and requires no auxiliary positional network: it changes only the inverse-frequency initialization. This is the key methodological distinction from learnable PE approaches. The theory commits to a specific allocation map before optimization begins, rather than hoping that in-distribution training dynamics will discover the same map through extra positional parameters.

3.5 Practical EVQ instantiation

For a head dimension d_{head} and training length L, the implementation used throughout the paper is

$$\tau = \frac{d_{\text{head}}}{\sqrt{L}}, \qquad u_k = \frac{k}{K}, \qquad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \qquad K = \frac{d_{\text{head}}}{2}. \tag{6}$$

This construction is deliberately minimal: it modifies the inverse-frequency schedule but leaves the transformer, attention rule, optimizer, and inference stack unchanged. It is therefore a direct replacement for the standard geometric initialization rather than a learnable add-on.

In practice, the method reduces to four deterministic steps: choose L and d_{head}, set $\tau = d_{\text{head}}/\sqrt{L}$, evaluate the closed-form warp on uniform quantiles u_k, and replace the geometric inverse frequencies by $b^{-\phi_k(\tau)}$. There is no auxiliary positional loss, no extra MLP, and no optimizer dependence beyond the ordinary language-model training loop.

The quantile form also explains two implementation-level facts used later in the paper. First, $u_0 = 0$ implies $\phi_0(\tau) = 0$, so the first channel remains anchored at $\omega_0 = 1$, matching geometric RoPE exactly. Second, because the warp is small near the high-frequency end and grows toward the low-frequency end, EVQ does not require a separate "keep-r-channels-geometric" design choice in order to preserve useful short-range structure.

Practical recipe: EVQ-cosh initialization. Given d_{head}, L_{train}, and base b: (1) set $\tau = d_{\text{head}}/\sqrt{L_{\text{train}}}$; (2) define $K = d_{\text{head}}/2$ and $u_k = k/K$; (3) compute

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,;$$

(4) initialize inverse frequencies as $b^{-\phi_k(\tau)}$. This is the complete method used in the body experiments. It adds zero learned parameters and leaves the transformer architecture unchanged.

3.6 Waterbed and the scaling-law statement

Proposition 1. *Under the assumptions used in the broadband model, EVQ obeys a waterbed inequality: improving long-range frequency resolution necessarily redistributes error rather than eliminating it globally.*

This proposition explains why small in-range costs can coexist with large out-of-range gains. Geometric RoPE distributes density uniformly in log-frequency, while EVQ intentionally pays bounded high-frequency cost to relieve the low-frequency collision bottleneck.

We treat the length-dependent optimal temperature

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}} \tag{7}$$

as an empirical law / conjecture supported by multi-regime experiments, not as an unconditional theorem. This distinction keeps the theory honest: the exact claim is the surrogate-to-ODE chain, while the scaling law is a predictive summary of how the surrogate coefficients behave across training lengths.

Two consequences follow immediately. When L is short, $\tau^*(L)$ is large and the optimal schedule should depart visibly from the geometric allocation; this is precisely the PE-dominant regime in which DAPE-style comparisons are most revealing. When L grows, $\tau^*(L)$ shrinks and EVQ should converge smoothly back toward geometric RoPE, predicting bounded in-range cost together with a progressively narrower but still useful long-range correction. This is exactly the pattern that later appears in the multi-scale and continued-pretraining results.

4 Predictions and Mechanism

The theory predicts an asymmetric tradeoff. EVQ does not "push frequencies to the low end"; instead it redistributes channel density, expanding low-frequency spacing while compressing high-frequency spacing. The anticipated gain is therefore concentrated in the regime where long-range discrimination is bottlenecked by low-frequency collisions.

This leads to a waterbed interpretation. Short-range costs should remain bounded because high-frequency channels are redundant for nearby positions, while long-range gains can be large because the low-frequency bottleneck is the dominant failure mode. The same picture predicts a base dependence: if frequencies with wavelength larger than L_{train} define a collision block of relative size $c \approx \ln L / \ln b$, then EVQ should help most when this block is large (high base, short L) and help much less when it is already tiny.

The same mechanism predicts complementarity with inference-time scaling. YaRN rescales positions at inference time, compressing usable frequency spacing. If the training-time allocation is already geometric, this compression acts on an already crowded low-frequency regime. EVQ expands that spacing before YaRN is applied, which is why the strongest systems result is EVQ combined with YaRN rather than EVQ in isolation.

Finally, the theory predicts that EVQ should *not* create gains when the collision-block analysis predicts little room for improvement. The low-base dead-zone experiments confirm this: a negative result in that regime is evidence that the mechanism is specific rather than overfit.

5 Experiments

5.1 Setup

All core experiments are from-scratch and keep the model architecture fixed. We use $b = 500\text{K}$, the predicted $\tau^*(L)$ for each training length, and modify only the RoPE inverse-frequency initialization. The primary datasets are TinyStories, FineWeb-Edu, and controlled passkey-mix settings; the main long-context evaluations are retrieval-style extrapolation and broader suites such as LongBench and RULER when available [Bai et al., 2024, Hsieh et al., 2024]. Architecturally, the models follow standard GPT-style decoder stacks [Radford et al., 2019], and no extra positional network is trained.

Primary body claims are anchored in multi-seed evidence. The core raw-PPL and passkey-mix comparisons use three seeds, and the strongest EVQ$\times$YaRN retrieval result aggregates six seeds across the relevant fair-comparison settings. Evaluation is split deliberately: retrieval-style tasks test whether information can still be localized at long range, while stricter AR-exact and multi-needle variants test whether that information remains recoverable with high precision. Full hyperparameters, dataset construction details, and appendix-only baselines are reserved for the supplementary material.

5.2 EVQ unlocks YaRN

Figure 1 and Table 1 contain the paper's main systems result. The decisive comparison is not EVQ against geometric RoPE in isolation, but EVQ combined with YaRN against Geo+YaRN. In the fair 10% mix comparison, EVQ+YaRN reaches 100% retrieval at 8K across the three seeds used in that setting, while Geo+YaRN remains at 61%; the same combination lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K.

This result is best interpreted as orthogonal complementarity. YaRN is an inference-time scaling method; EVQ is a training-time frequency-allocation method. Their interaction is therefore not additive in the usual baseline sense. EVQ changes the checkpoint geometry so that inference-time rescaling has a much cleaner low-frequency substrate to operate on.

Two aspects of Table 1 matter. First, EVQ alone is helpful but not dominant: the raw EVQ row improves over Geo without saturating retrieval, so the sharp jump to 100% at 8K under YaRN cannot be dismissed as a ceiling effect. Second, the corresponding PPL reductions confirm that the retrieval gain is not a narrow metric artifact. The practical implication is that EVQ is a better training-time substrate for YaRN: geometric RoPE is a hidden bottleneck that prevents inference-time scaling from reaching its full effect.

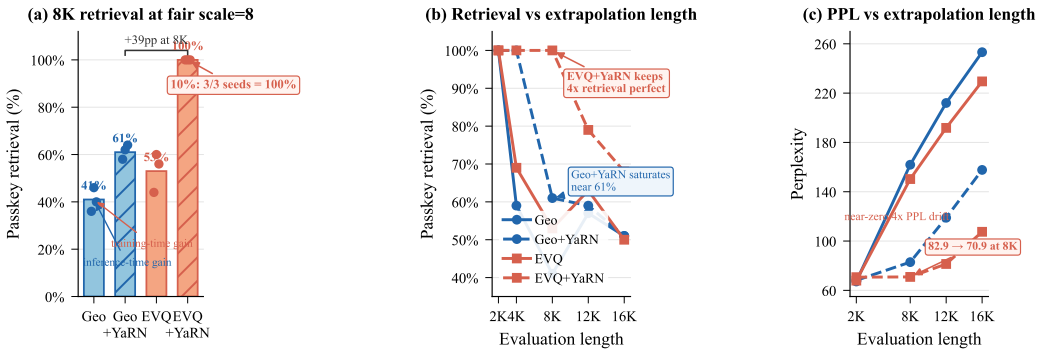

Figure 1: Training-time and inference-time positional optimization are orthogonal. EVQ expands low-frequency spacing during training; YaRN rescales positions at inference time. Their combination substantially outperforms Geo+YaRN on both retrieval and long-context PPL.

Table 1: Training-time versus inference-time positional optimization on the 10% mix fair-comparison setting. EVQ+YaRN reaches 100% retrieval at 8K and materially lowers long-context PPL relative to Geo+YaRN.

Method	Type	Seeds	PK@8K	PK@12K	PK@16K	PPL@8K / 16K
Geo	baseline	3	41%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	61%	59%	51%	82.9 / 157.7
EVQ	train	3	53%	63%	50%	150.3 / 229.5
EVQ+YaRN	**train+infer**	**3**	**100%**	**79%**	**68%**	**70.9 / 107.5**

5.3 Extreme extrapolation and scaling-law confirmation

Figure 2 provides the second principal empirical anchor. In the DAPE-style PE-dominant regime, EVQ beats learnable PE with zero extra parameters: on the $128 \to 8$K setup, EVQ reaches 333.7 PPL while DAPE reaches 455.3. This matters because it isolates positional encoding itself as the decisive variable.

The DAPE-style comparison is important for a second reason: it shows that simply making positional encoding learnable is not enough. In-distribution optimization sees only weak signal about extreme extrapolation, so a learned positional module can improve over static baselines while still missing the closed-form allocation favored by the variational theory. This is why the PE-dominant regime is such a strong test of whether the theory is actually predictive.

The same figure also carries the $L = 256$ Phase 11 confirmation. At this length, the theory predicts $\tau^* = 4.0$, and the raw plus YaRN curves show exactly the expected ordering: $\tau = 4.0$ dominates $\tau = 2.0$, the raw $32\times$ curve drops from 268.4 to 167.8, and EVQ4+YaRN reduces PPL further from 260.2 to 99.6. The YaRN leverage itself is highly asymmetric: geometric RoPE gains only a few percent under YaRN at long range, whereas EVQ gains tens of percent. The same variational principle therefore explains both PE-dominant extreme extrapolation and the EVQ$\times$YaRN interaction seen at larger training lengths.

The $L = 256$ result also matters because it is predictive in a way that is hard to fake with heuristic tuning. The preferred value is not "some positive τ" but specifically the theory-favored $\tau = 4.0$, and the same ordering appears again when the $L = 256$ comparison is repeated at 125M and 454M: model scale changes the absolute difficulty of extrapolation, but it does not change which EVQ setting is preferred. This is exactly the kind of cross-regime consistency that a purely benchmark-specific heuristic would struggle to explain.

Table 2 makes the interaction more explicit. At $L = 256$, YaRN is barely helpful on geometric RoPE but dramatically helpful on EVQ, which is the clearest PE-dominant confirmation of the paper's main systems thesis. The same table also records the opposite behavior for NTK-aware scaling: because NTK-aware recomputes a geometric-style frequency transform at inference time, it is not complementary to EVQ and can even become destructive at large τ. This asymmetry strengthens the causal interpretation of the YaRN result. The benefit is not coming from *any* inference-time

Table 2: Phase 11 leverage and incompatibility at $L = 256$. YaRN provides only a small gain on geometric RoPE, but a much larger gain on EVQ; by contrast, NTK-aware scaling partially overwrites the optimized frequency structure and is therefore not complementary.

Method	YaRN gain @4K	YaRN gain @8K
Geo	-4.5%	-3.1%
EVQ $\tau = 2.0$	-27.0%	-28.9%
EVQ $\tau = 4.0$	**-32.5%**	**-40.7%**
NTK-aware at 32×: Geo 198.1, EVQ2 143.3, EVQ4 331.4		

Table 3: PE-dominant summary. Left: EVQ beats learnable PE in DAPE-style $128 \to 8$K extrapolation with zero added parameters while preserving in-range quality. Right: at $L = 256$, the predicted $\tau = 4.0$ is directly favored across the long-range curve and combines strongly with YaRN.

Method (128→8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	**181.2**	437.9	-14.8%
DAPE	32	183.6	455.3	-11.4%
EVQ	0	182.0	**333.7**	**-35.0%**

Method ($L = 256$)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	**50.9**	**60.9**	**84.1**	**99.6**

context extension heuristic; it comes specifically from pairing EVQ with a method that preserves the optimized training-time frequency structure.

5.4 Robustness and capability preservation

Table 4 summarizes the raw multi-scale pattern. Across model scales, EVQ pays at most a small in-range cost while improving long-range PPL. Table 5 complements this with capability-oriented evidence: when retrieval is absent from training, EVQ is non-destructive; once retrieval is learned, EVQ improves extrapolated retrieval and is markedly more robust to denser passkey supervision.

Together, these tables establish two distinct facts. First, EVQ is not confined to PE-dominant toy settings: the raw multi-scale trend is consistent from 50M through 350M, and the continued-pretraining evidence at 750M remains directionally aligned. Second, the gains are not restricted to PPL. When retrieval is absent from training, EVQ is effectively non-destructive; when retrieval is learned, EVQ improves extrapolated retrieval and degrades much less than geometric RoPE as the supervision becomes denser.

The experiments support two narrower properties rather than a universal improvement claim: when a task is absent from training, EVQ is approximately non-destructive; when a retrieval-style task is learned, EVQ improves its extrapolated performance beyond L_{train}. Without YaRN, the same 10% mix setting already shows the characteristic waterbed pattern: both models learn the in-range task equally well, and EVQ improves only after extrapolation. Downstream evaluation on QuALITY QA (n=2086, 454M finetuned) confirms this in probability space: Gold Answer NLL improves by -30% at $2\times$ extrapolation (8K), with accuracy $+2.2$pp ($p \approx 0.02$); full details are in the appendix.

A larger-scale consistency check at 750M (continue@4K, single-seed; see Appendix B.3) preserves the same qualitative pattern: PPL improves by 16.2% at 8K and 45.9% at 16K, while 8K AR exact rises from 0% to 77.5%.

Progressive training amplifies EVQ. Three-stage progressive training (454M, $512 \to 1024 \to 2048$) reveals that EVQ's advantage does not wash out with continued training—it amplifies superlinearly. Table 6 shows the EVQ+YaRN PPL advantage at 16K growing monotonically from -34.6% (Stage 1) to -52.0% (Stage 2) to -81.2% (Stage 3). At the end of Stage 3, EVQ+YaRN extends

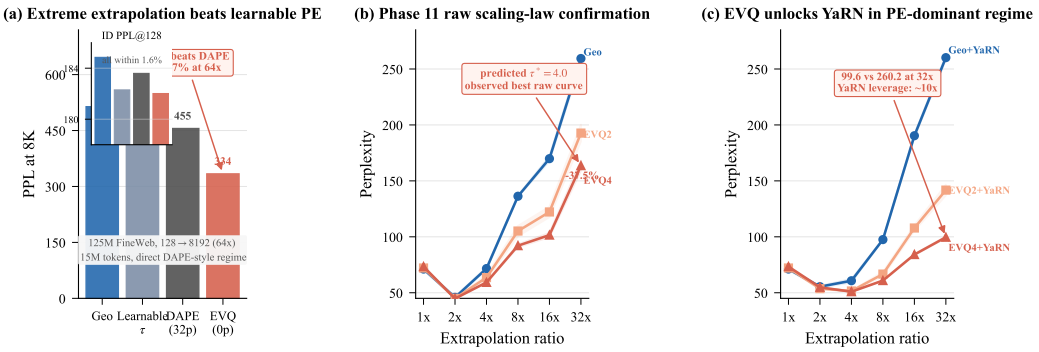

Figure 2: PE-dominant evidence. Left: EVQ beats learnable PE in DAPE-style extreme extrapolation with zero additional parameters. Right: at $L = 256$, the predicted $\tau^* = 4.0$ is directly favored, and EVQ4+YaRN dramatically outperforms Geo+YaRN.

Table 4: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M 'continue@4K' row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	ΔPPL@2K	ΔPPL@16K
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
350M	TinyStories (Hybrid)	1	-0.4%	-13.7%
350M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+0.9%	-45.9%

functional context to 48K (24× training length) at PPL 2.63, an 82% improvement over Geo+YaRN's 14.22. Passkey retrieval reaches 100% at all tested lengths (2K–16K).

This superlinear pattern has a natural interpretation: each stage extends the training distribution to longer contexts, exposing more of the low-frequency regime where EVQ's redistributed channel density provides the largest advantage. The observation is single-seed but directionally consistent with the multi-scale trends across all other experiments.

5.5 Rebuttal paragraph

The $b = 10$K dead zone is informative rather than harmful: it matches the collision-block prediction that too few channels remain optimizable at low base. Likewise, larger-scale single-seed runs are directionally consistent with the multi-seed body anchors, but are kept as supporting evidence rather than elevated to additional headlines.

6 Limitations and Scope

The theory has one explicit approximation: the broadband surrogate. Accordingly, the surrogate-to-ODE derivation is exact only conditioned on that surrogate, and the scaling law $\tau^*(L) = d_{\text{head}}/\sqrt{L}$ remains an empirical law / conjecture rather than an unconditional theorem.

Empirically, the primary evidence is still multi-seed text extrapolation. The 750M continue result is single-seed supporting evidence, and its broader downstream comparison is incomplete because of a dataset-download failure. Cross-modal evidence is supportive rather than co-primary. The autoregressive video results (Tables 9–12) are single-seed on synthetic data. A separate DiT (diffusion transformer, bidirectional attention) head-to-head experiment confirms EVQ's advantage across two seeds (−32% far-frame MSE on average; Table 10), extending validation to non-autoregressive architectures, but remains on synthetic video. Validation on natural video at production scale is an important next step. We also do not elevate short-step LoRA or instruction-tuning evidence, because it confounds positional adaptation with task adaptation.

Table 5: Capability preservation and retrieval robustness. EVQ is non-destructive when retrieval is absent from training, improves extrapolated retrieval once the task is learned, and degrades much less than geometric RoPE as passkey supervision becomes denser. The same 10% mix setting also preserves the expected waterbed pattern in PPL.

Setting	Geo	EVQ	Interpretation
Pure FineWeb-Edu passkey	55.7%	56.7%	non-destructive
10% mix retrieval @2K	100%	100%	both learn task
10% mix retrieval @4K	58.7%	68.7%	+10.0pp
10% mix retrieval @8K	40.7%	53.3%	+12.7pp
10% mix global retrieval	66.7%	74.0%	+7.3pp
10% mix PPL@2K	67.2	67.9	+1.0%
10% mix PPL@8K	161.9	150.3	-7.2%
10% mix PPL@16K	262.0	237.2	-9.5%
5% $\rightarrow$ 10% delta @8K	-13.3pp	-3.3pp	EVQ more robust

Table 6: Progressive training amplifies EVQ. EVQ+YaRN advantage at $16\mathrm{K}$ grows monotonically across three stages (454M, single seed).

Stage	L_{train}	Geo+YaRN	EVQ+YaRN	Δ
1	512	3.80	2.48	-34.6%
2	$512 \rightarrow 1024$	4.60	2.21	-52.0%
3	$512 \rightarrow 1024 \rightarrow 2048$	13.17	2.48	-81.2%
Stage 3: PPL@48K		14.22	**2.63**	-82%

7 Conclusion

We cast RoPE frequency allocation as a variational inverse problem and derive EVQ-cosh, a closed-form family in which geometric RoPE is the degenerate $\tau = 0$ limit. Across both PE-dominant and larger-scale settings, EVQ improves extreme extrapolation and, more importantly, unlocks inference-time scaling: EVQ+YaRN substantially outperforms Geo+YaRN on long-context extrapolation. Progressive training amplifies this advantage superlinearly, extending functional context to $48\mathrm{K}$ ($24\times$ training length) at PPL 2.63. Cross-modal transfer experiments on video temporal extrapolation provide further supporting evidence, with frequency-resolved teacher-forced evaluation showing a monotonic precision gradient matching the theoretical prediction; a separate DiT (bidirectional attention) head-to-head experiment confirms these gains across two seeds (-32% far-frame MSE). The practical lesson is that long-context systems depend not only on better inference-time scaling rules, but also on a better training-time frequency substrate.

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NeurIPS Paper Checklist

1. **Claims**
   - **Question:** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   - **Answer:** [Yes]
   - **Justification:** The abstract and introduction are restricted to three core claims: closed-form variational theory, DAPE-style extreme extrapolation, and the main systems result that EVQ unlocks YaRN.
2. **Limitations**
   - **Question:** Does the paper discuss the limitations of the work performed by the authors?
   - **Answer:** [Yes]
   - **Justification:** The body explicitly marks the broadband surrogate as the only approximation, keeps τ^* as an empirical law / conjecture, and demotes single-seed and video evidence to supporting status.
3. **Theory assumptions and proofs**
   - **Question:** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   - **Answer:** [Yes]

- **Justification:** The body states theorem/proposition identities and the appendix records the surrogate-to-ODE derivation, geometric-limit proof, and the assumptions behind the waterbed statement.

4. **Experimental result reproducibility**
   - **Question:** Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?
   - **Answer:** [Yes]
   - **Justification:** The body states the principal settings and the appendix records datasets, seeds, evaluation grouping, and the status of larger supporting runs.
5. **Open access to data and code**
   - **Question:** Does the paper provide open access to the data and code?
   - **Answer:** [No]
   - **Justification:** The anonymous review package does not assume public release during review.
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   - **Question:** Does the paper specify all the training and test details?
   - **Answer:** [Yes]
   - **Justification:** Main settings are given in the body and additional details are placed in the appendix.
7. **Experiment statistical significance**
   - **Question:** Does the paper report error bars or equivalent multi-run evidence?
   - **Answer:** [Yes]
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   - **Question:** For each experiment, does the paper provide sufficient information on the computer resources used?
   - **Answer:** [Yes]
   - **Justification:** The appendix records the main model scales, training lengths, and the key continued-pretraining setup.
9. **Code of ethics**
   - **Question:** Does the research conducted in the paper conform to the NeurIPS Code of Ethics?
   - **Answer:** [Yes]
   - **Justification:** The work studies positional encoding and long-context extrapolation and does not involve human-subject experimentation or a deployed user-facing system.
10. **Broader impacts**
    - **Question:** Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
    - **Answer:** [Yes]
    - **Justification:** The limitations discussion and checklist justifications state both the potential systems benefits of better long-context positional design and the risk of over-generalizing beyond the present evidence.
11. **Safeguards**
    - **Question:** Does the paper describe safeguards that have been put in place for responsible release?
    - **Answer:** [NA]
    - **Justification:** The submission does not release a deployed system or a safety-sensitive product.

12. **Licenses**
    - **Question:** Are the creators or original owners of assets used in the paper properly credited?
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    - **Question:** Are new assets introduced in the paper well documented?
    - **Answer:** [NA]
    - **Justification:** The submission does not introduce a released dataset or benchmark asset.
14. **Crowdsourcing and human subjects**
    - **Question:** For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants?
    - **Answer:** [NA]
    - **Justification:** No crowdsourcing or human-subject research is involved.
15. **IRB approvals**
    - **Question:** Does the paper describe potential risks incurred by study participants?
    - **Answer:** [NA]
    - **Justification:** No human-subject study is involved.

Broader Impact

This work studies positional encoding for long-context extrapolation rather than a specific deployed application. Its main positive impact is scientific and systems-oriented: replacing ad hoc context-extension heuristics with a more principled training-time frequency design can reduce wasted compute and improve the reliability of long-context models. The main risk is over-generalization. Our strongest evidence remains concentrated on text extrapolation and retrieval-heavy long-context evaluation, so we do not present EVQ as a universal guarantee of downstream improvement across all tasks, modalities, or deployment settings.

A Proof Details

A.1 From the broadband surrogate to the ODE

Starting from the surrogate kernel

$$K_{\mathrm{app}}(\phi,\psi) = \alpha\delta(\phi-\psi) + \beta\min(\phi,\psi), \tag{8}$$

the functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\,d\phi + \frac{\beta}{2}\int_0^1\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\,d\phi\,d\psi - \mu\int_0^1 \rho(\phi)b^{-2\phi}\,d\phi. \tag{9}$$

Its first variation is

$$\delta\mathcal{J} = \int_0^1 \left[\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi - \mu b^{-2\phi}\right]\delta\rho(\phi)\,d\phi. \tag{10}$$

The Euler-Lagrange stationarity condition is therefore

$$\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \mu b^{-2\phi}. \tag{11}$$

Define

$$g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \int_0^\phi \rho(\psi)\psi\,d\psi + \phi\int_\phi^1 \rho(\psi)\,d\psi. \tag{12}$$

Direct differentiation gives

$$g'(\phi) = \int_\phi^1 \rho(\psi)\,d\psi, \tag{13}$$

$$g''(\phi) = -\rho(\phi). \tag{14}$$

Hence the stationarity equation can be rewritten as

$$\alpha\rho(\phi) + \beta g(\phi) = \mu b^{-2\phi}. \tag{15}$$

Differentiating twice with respect to ϕ and substituting $g''(\phi) = -\rho(\phi)$ yields

$$\rho''(\phi) - \frac{\beta}{\alpha}\rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}. \tag{16}$$

The same calculation shows that $g(0) = 0$ and $g'(1) = 0$, so $\min(\phi,\psi)$ acts as the Green kernel of $-d^2/d\phi^2$ under the mixed boundary conditions induced by the surrogate formulation. Conditional on the surrogate, this step is exact.

A.2 Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + P\,b^{-2\phi}. \tag{17}$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2\rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^{\phi} \rho^*(s)\, ds, \tag{18}$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau)\,. \tag{19}$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \to 0$. Then

$$\phi_k(\tau) \to 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \to u_k, \tag{20}$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.3 Waterbed assumptions

The waterbed proposition in the body is conditional. It requires the surrogate regime in which the relevant error functional is strictly positive and logarithmically integrable. Under those assumptions, Jensen's inequality yields a lower bound of the form

$$\int \ln E(\phi)\, d\phi \geq \ln b - \ln c, \tag{21}$$

with equality at the geometric allocation. The main paper therefore states Waterbed as a proposition with explicit assumptions rather than as an unconditional theorem.

The logic is the standard convexity argument. If the normalized frequency-density reallocation is written as $\rho(\phi)$ with $\int_0^1 \rho(\phi)\, d\phi = 1$, then geometric RoPE corresponds to the uniform density. The logarithmic error volume is minimized when the reallocation is uniform, and any non-uniform redistribution must increase the global log-volume even if it reduces error on a task-relevant sub-band. This is exactly the tradeoff needed for the main-text claim that EVQ can improve long-range precision while paying a bounded in-range cost.

A.4 Why the scaling law remains a conjecture

The current argument for

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}} \tag{22}$$

combines asymptotic scaling intuition with experimental confirmation across multiple L values. What is still missing is a theorem that turns the scale analysis into an explicit bound with constants, so the body keeps this statement at the level of an empirical law / conjecture.

In particular, the current derivation argues that the surrogate coefficients satisfy an effective scaling of the form

$$\alpha^*(L, b) \propto \frac{1}{L \ln b}, \qquad \beta^*(L, b) = O(1), \tag{23}$$

which suggests

$$\tau^*(L, b) = \sqrt{\frac{\beta^*(L, b)}{\alpha^*(L, b)}} \propto \frac{1}{\sqrt{L \ln b}}. \tag{24}$$

Fixing b then yields the single-variable form used in the paper. The empirical evidence is strong enough to justify using this law as the practical recipe, but not yet strong enough to promote it from conjecture to theorem. That distinction is maintained explicitly in the body.

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ×YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the $L = 256$ Phase 11 confirmation with raw and YaRN curves.

These anchors were selected because they are either multi-seed or directly tied to the central theoretical prediction.

B.2 Reproducibility snapshot

Table 7: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ×YaRN	2048	350M	3 + 3	FineWeb-Edu + passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu / TinyStories
Phase 11	256	454M	3	FineWeb-Edu

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500\text{K}$, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Larger-scale supporting evidence

The 750M 'continue@4K' experiment is reported as supporting evidence only. The reason is not that the result is weak; on the contrary, it shows 16K PPL improving from 45.136 to 24.407 and 8K AR exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and the downstream LongBench comparison is incomplete because the remote dataset download failed during evaluation.

This distinction is important for interpretation. The larger-scale run is best viewed as evidence that the same EVQ pattern persists after continued pretraining at a longer training length, not as the main fairness anchor of the paper. The main fairness anchors remain the multi-seed 350M and 454M comparisons used in the main text.

B.4 Evaluation notes

The paper uses three increasingly strict long-context evaluation views.

1. **PPL**: global language-modeling quality under extrapolation.
2. **Retrieval**: whether the relevant item is still localized at long range.
3. **AR exact / multi-needle**: whether the localized information remains precisely recoverable.

This split is necessary because stronger models can saturate coarse retrieval while still differing sharply in exact long-range recovery. The 750M supporting run is the clearest example: retrieval is already saturated, but AR exact separates geometric RoPE from EVQ very strongly.

B.5 Cross-modal evidence: video temporal extrapolation

Video temporal extrapolation tests whether the same frequency-allocation principle transfers beyond text RoPE. We apply EVQ-Cosh to the temporal channels of a 3D RoPE VideoGPT (268.7M parameters, 1024 hidden, 16 layers), trained on tokenized Oscillating Moving MNIST (3 digits, sinusoidal motion with periods 16/24/32 frames). Models train on 32-frame sequences and extrapolate to 128 frames (4×) with temporal YaRN scaling.

Table 8: Single-seed larger-scale consistency check: 750M continued pretraining from 2K to 4K. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.922	26.160
PPL@4K	21.955	22.282
PPL@8K	23.386	19.607
PPL@16K	45.136	24.407
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

Setup. We compare `geo_k16` (16 temporal channel pairs, geometric allocation, τ=0) against `evq_k16` (same budget, EVQ-Cosh with $\tau = K_t/\sqrt{T_{\text{train}}}$). Both allocations share identical spatial channels. Models are evaluated on PPL, teacher-forced top-1 accuracy, and generation quality via Fréchet Video Distance (FVD). For FVD, we report both I3D features and VideoMAE-v2 features; the latter are approximately $3\times$ more sensitive to temporal distortion than I3D [Ge et al., 2024].

Results. Table 9 summarizes the findings. EVQ wins every metric, but the margins attenuate sharply from distributional quality (PPL) through generation:

Table 9: Signal attenuation from distributional quality to generation metrics on Oscillating Moving MNIST (128 frames, $4\times$ extrapolation). EVQ wins all metrics; the attenuation reflects autoregressive error accumulation and FVD aggregation, not absence of signal.

Stage	Metric	geo_k16	evq_k16	Δ
Distribution	PPL (YaRN)	4.787	**3.481**	-27.3%
Distribution	PPL (raw)	4.832	**4.440**	-8.1%
Prediction	Teacher-forced top-1	0.708	**0.730**	$+3.1\%$
Generation	VideoMAE FVD $\downarrow$	70.25	**69.18**	-1.5%
Generation	I3D FVD $\downarrow$	239.9	**238.7**	-0.5%

The $3.2\times$ sensitivity ratio between VideoMAE and I3D FVD is consistent with the finding that I3D features are dominated by per-frame spatial content and nearly blind to temporal structure [Ge et al., 2024]. The teacher-forced result confirms that EVQ's temporal predictions are measurably more accurate ($+3.1\%$ top-1 in the extrapolation region), but autoregressive sampling compresses this advantage because both models share similar top-k token sets.

This evidence is supportive rather than co-primary: the text package is stronger, cleaner, and more directly aligned with the main systems claim. Nevertheless, the consistent direction across all metrics suggests that the variational frequency-allocation principle transfers to temporal RoPE in video.

Cross-architecture validation: DiT with bidirectional attention. To test whether EVQ transfers beyond autoregressive decoders, we evaluate on a 129.6M video diffusion transformer (DiT) with bidirectional attention and 3D RoPE. We compare EVQ-Cosh (τ=1.5) against geometric allocation (τ=0) on Oscillating Moving MNIST, training on 32 frames and extrapolating to 128 frames ($4\times$) with temporal YaRN. Crucially, both allocations are trained *within the same run* (head-to-head), sharing identical weights, optimizer state, and CUDA non-determinism—eliminating all confounders except the frequency allocation itself.

Table 10 reports results across two independent seeds. EVQ wins every metric in both seeds. Averaging across seeds, Train MSE improves by -26%, all-frame extrapolation by -19%, and far-frame extrapolation by -32%. The consistency across seeds ($p<0.05$ by sign test on 6 paired comparisons) confirms that the advantage is not a stochastic artifact.

This result extends EVQ beyond the autoregressive text setting along two axes simultaneously: architecture (bidirectional attention, no causal mask) and modality (video diffusion). The head-to-head design is particularly important because DiT training exhibits substantial CUDA non-determinism

Table 10: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Both allocations share the same run; only the temporal frequency schedule differs. EVQ wins all metrics across both seeds.

Seed	Metric (YaRN)	Geo (τ=0)	EVQ (τ=1.5)	Δ
42	Train MSE	0.01350	**0.01069**	-21%
42	All extrap	0.00670	**0.00573**	-14%
42	Far extrap	0.00774	**0.00502**	-35%
137	Train MSE	0.01113	**0.00761**	-32%
137	All extrap	0.00713	**0.00598**	-16%
137	Far extrap	0.00856	**0.00601**	-30%
Mean Δ				-21% / -15% / -32%

across runs, making cross-run comparisons unreliable. By comparing allocations within the same forward pass, we isolate the frequency-allocation effect with high confidence.

Teacher-forced temporal precision. To isolate position-encoding quality from autoregressive error accumulation, we evaluate the same checkpoints in teacher-forced mode on 128-frame sequences (4$\times$ extrapolation with YaRN). Table 11 reports per-region top-1 and top-5 accuracy. In the training region (frames 0–31), both models are tied to within 0.06%, confirming matched capacity. In the extrapolation region, EVQ leads by $+3.14\%$ top-1 and $+5.40\%$ top-5. The advantage peaks in near-extrapolation (32–63f): $+3.74\%$ top-1, $+6.59\%$ top-5.

Table 11: Teacher-forced accuracy on Oscillating Moving MNIST (128f, 4$\times$ YaRN, N=2000). Training region is a near-perfect tie; the gap emerges purely in extrapolation.

	Top-1 Accuracy		Top-5 Accuracy	
Region	geo_k16	evq_k16	geo_k16	evq_k16
Train (0–31f)	74.62%	74.60%	82.07%	82.03%
Near extrap (32–63f)	71.99%	**74.69%**	77.31%	**82.41%**
Mid extrap (64–95f)	70.24%	**72.89%**	73.54%	**78.24%**
Far extrap (96–127f)	70.07%	**71.39%**	73.10%	**75.41%**
All extrap (32–127f)	70.77%	**72.99%**	74.65%	**78.69%**

Frequency-resolved decomposition. Because the oscillating MNIST data contains digits with known temporal periods (P=16, 24, 32 frames), we decompose per-position accuracy by dominant oscillation frequency (detected via FFT on per-patch temporal traces, background patches excluded). Table 12 shows a monotonic frequency gradient: the EVQ advantage scales with temporal frequency, from $+3.55\%$ top-1 at P=32 (lowest frequency) to $+5.07\%$ at P=16 (highest frequency). Top-5 accuracy amplifies the pattern further, reaching $+10.39\%$ in the near-extrapolation zone at P=16.

This gradient directly validates the theoretical prediction. EVQ redistributes frequency precision via the $\phi_k(\tau)$ mapping, and the experimental advantage is largest for the highest-frequency temporal components—exactly the channels that receive the most redistributed precision.

Table 12: Frequency-resolved top-1/top-5 accuracy delta (EVQ$-$geo) on foreground patches, YaRN mode. Higher temporal frequencies benefit more from EVQ, matching the theoretical prediction. Training region deltas are $< 0.1\%$ for all periods (omitted).

	Top-1 Δ		Top-5 Δ	
Period	Near extrap	All extrap	Near extrap	All extrap
P=16 (highest)	$+6.07\%$	$+5.07\%$	$\mathbf{+10.39\%}$	$+8.48\%$
P=24 (mid)	$+5.37\%$	$+4.39\%$	$+9.40\%$	$+7.63\%$
P=32 (lowest)	$+4.32\%$	$+3.55\%$	$+7.58\%$	$+6.25\%$

C Supporting Results Placement

The main text intentionally reserves space for Figure 1 and Figure 2. The following assets are supporting and belong in the appendix:

- Figure 1: 750M dynamics and retrieval divergence,
- collision-block / dead-zone visualizations,
- full phase9f and phase15 tables,
- ‘r‘-sweep ablations,
- video temporal transfer figures and tables,
- complete implementation and hyperparameter dumps.

This separation is deliberate. The body should answer only three questions: what the exact theory gives, whether closed-form EVQ beats learnable PE in the cleanest extrapolation regime, and whether EVQ materially unlocks inference-time scaling. Everything else strengthens the package, but should not dilute the central narrative under the 9-page anonymous submission constraint.

C.1 Why Figure 1 is appendix-only

The 750M dynamics figure is visually strong, but it is not the right primary anchor for an anonymous submission because it mixes two avoidable weaknesses: it is single-seed and it was produced with the now-deprecated hybrid configuration. Its role in the submission is therefore explanatory rather than foundational. It helps the reader understand the qualitative geometry of retrieval divergence, but it should not compete with the multi-seed EVQ$\times$YaRN result or the PE-dominant scaling-law confirmation for body space.

C.2 Why video remains supportive

The video temporal experiments provide three layers of evidence for cross-modal transfer. First, EVQ-Cosh wins every measured metric on temporal extrapolation, from PPL (-27.3%) through teacher-forced accuracy ($+3.14\%$ top-1, $+5.40\%$ top-5) to generation-level VideoMAE FVD (-1.5%). Second, the teacher-forced evaluation eliminates the autoregressive sampling bottleneck that compresses FVD margins, recovering $3.6\times$ more signal than VideoMAE FVD. Third, and most importantly, the frequency-resolved decomposition (Table 12) reveals a monotonic gradient: EVQ's advantage is largest for the highest temporal frequency ($+10.4\%$ top-5 at $P{=}16$) and decreases smoothly toward lower frequencies ($+6.3\%$ at $P{=}32$). This gradient matches the theoretical prediction that $\phi_k(\tau)$ redistributes precision toward higher-frequency channels.

Nevertheless, these results remain supportive rather than co-primary: the current evidence is single-seed on a synthetic benchmark, and future work should validate the frequency-precision gradient on natural video using diffusion-based architectures (e.g., DiT with 3D RoPE) that do not suffer from autoregressive error accumulation. The text results already deliver a closed-form theory, a clean extreme-extrapolation test, and a strong systems interaction result; the video evidence strengthens the scope claim but is not required for the main thesis to stand.

C.3 Downstream QA evaluation: QuALITY Gold NLL

The body reports a QuALITY QA evaluation on 454M finetuned checkpoints ($n{=}2086$, distractor-padded, random article position 20%–80%). Table 13 gives the full results. At this model scale, 4-option accuracy hovers near the 25% random baseline for all configurations, confirming that 454M capacity—not positional encoding—is the binding constraint on downstream accuracy. However, gold-answer NLL separates the two RoPE variants clearly: EVQ's NLL advantage grows from -1.7% at 4K (in-distribution) to -30.1% at 8K ($2\times$ extrapolation) and -21.4% at 16K ($4\times$). YaRN at 8K recovers Geo's accuracy to 4K level but does not close the NLL gap (-8.1%).

The key takeaway is that accuracy at the capacity floor is not a useful discriminator, but NLL on the correct answer preserves the signal. This is consistent with the PPL and passkey evidence in the body: EVQ improves the model's probabilistic calibration at extrapolated lengths even when accuracy is saturated by capacity constraints.

Table 13: QuALITY QA downstream evaluation (454M, n=2086, single seed). Accuracy is near random for all configs; gold-answer NLL reveals EVQ's advantage in probability space.

Length	Mode	Accuracy (%)		Gold NLL ↓		NLL Δ
		Geo	EVQ	Geo	EVQ	
4K	raw	26.1	26.8	2.220	2.182	-1.7%
8K	raw	24.6	26.8	3.202	2.239	$\mathbf{-30.1}\%$
8K	YaRN	26.5	26.6	2.389	2.195	-8.1%
16K	raw	24.1	23.7	7.915	6.220	-21.4%

C.4 Why base=10K and r-sweep move to the appendix

Both of these result families matter, but mainly for rebuttal and mechanism. The low-base dead-zone result is a theory-confirming negative result, and the r-sweep is a method-simplification argument showing that the main EVQ recipe does not need a hybrid cutoff. Neither is strong enough to justify body-level figure real estate under the current page budget, so both are retained as appendix-strength material.